Data Mining Assignment

Question 1: Applications of Statistical Learning

The assignment considers three major statistical learning approaches: classification, regression, and cluster analysis.

A. Classification Applications

Classification is used when the response variable belongs to a category or class. The task is to assign a new observation to one of the predefined classes.

Application	Response	Predictors	Goal	Reason
1. Email Spam Detection	Spam / Not spam	Word frequencies, phrases, sender address, number of links, capital letters, HTML content	Prediction	The objective is to correctly classify new incoming emails.
2. Medical Diagnosis	Disease / No disease	Age, blood pressure, test results, symptoms, cholesterol level	Inference	The goal may be to understand which factors are associated with the presence of disease; classification can also support prediction.
3. Telecom Customer Churn	Churn / Does not churn	Data usage, complaints, subscription type, payment history, call frequency	Prediction	The company wants to identify customers who are likely to leave so that preventive action can be taken.

B. Regression Applications

Regression is normally used when the response variable is quantitative or continuous. It can be used either to predict a numerical value or to understand relationships between variables.

Application	Response	Predictors	Goal	Reason
1. Predicting House Prices	Sale price	Square footage, bedrooms/bathrooms, lot size, neighborhood, house age, school district, distance to city center	Prediction	The purpose is to estimate the price of a particular house.
2. Employee Salary Analysis	Salary	Experience, education, job level, age, location	Inference	The objective is to understand how these factors are associated with salary.
3. Advertising Spend and Sales	Sales revenue	TV, radio, newspaper/online advertising budgets	Inference	The company wants to understand which advertising channels are related to sales and how strong those relationships are.

C. Cluster Analysis Applications

Cluster analysis is an unsupervised learning technique. There is no predefined response variable. The objective is to discover natural groups or patterns within the data.

1. **Market Segmentation**

   Customers can be grouped using purchasing behavior, demographics, browsing habits, and spending patterns. Possible groups include budget shoppers, luxury buyers, and occasional browsers.

2. **Gene Expression Analysis**

   Genes or patients can be grouped according to similarities in gene expression. Similar behavior may indicate shared biological functions or disease subtypes.

3. **Document/Topic Clustering**

   Documents can be grouped according to similarities in word usage and content. For example, articles can naturally form groups related to politics, sports, or technology.

Question 2: K-Nearest Neighbors (KNN)

K-Nearest Neighbors classifies a new observation by looking at nearby observations in the training data. The assignment uses the test point $(0, 0, 0)$ and Euclidean distance.

Euclidean Distance

For a test point at $(0, 0, 0)$, the distance is:

$$\text{Distance} = \sqrt{X_1^2 + X_2^2 + X_3^2}$$

Observation	Distance	Class
1	3.00	Red
2	2.00	Red
3	3.16	Red
4	2.24	Green
5	1.41	Green
6	1.73	Red

$K = 1$

When $K = 1$, KNN uses only the single closest observation. Observation 5 has the smallest distance, 1.41, and its class is Green. Therefore, the prediction is Green.

$K = 3$

When $K = 3$, the three closest observations are Observation 5 (Green), Observation 6 (Red), and Observation 2 (Red). Red has the majority with two observations, so the prediction is Red.

With K equal to 3, I consider the three nearest observations and use majority voting. Two are Red and one is Green, so the prediction changes to Red.

Choosing K for a Highly Nonlinear Decision Boundary

A small K is generally preferred when the true decision boundary is highly nonlinear because it allows KNN to follow complex local patterns. A large K produces a smoother boundary and may oversimplify the relationship.

However, a very small K can be sensitive to noise and therefore have high variance. In practice, K should be selected using cross-validation rather than simply choosing $K = 1$.

For a highly nonlinear boundary, I would generally prefer a smaller K because it gives the model more flexibility. However, I would use cross-validation to select the final value because a very small K can be sensitive to noise.

Question 3: Linear Model Versus Cubic Model

The assignment compares a linear regression model with a cubic regression model. The cubic model is more flexible because it includes additional terms.

$$\text{Linear: } Y = \beta_0 + \beta_1 X + \epsilon$$

$$\text{Cubic: } Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$$

(a) True Relationship Is Linear: Training RSS

The cubic model cannot have a higher training RSS than the linear model. The reason is that the cubic model contains the linear model as a special case.

If $\beta_2 = 0$ and $\beta_3 = 0$, the cubic model becomes the linear model. Therefore, the cubic model has all the options of the linear model plus additional flexibility. It can therefore achieve a training RSS that is equal to or lower than the linear model.

The cubic model cannot have a higher training RSS because it contains the linear model as a special case. A more flexible model has all the choices of the simpler model plus additional choices.

(b) True Relationship Is Linear: Test RSS

For test RSS, the answer is not automatic. Although the cubic model can fit the training data more closely, its additional flexibility can cause it to fit random noise. This is overfitting.

Because the true relationship is linear, the simpler linear model may generalize better. Without additional information, we cannot determine which model will have the lower test RSS.

Training performance and test performance are different. A cubic model may fit the training data better but still perform worse on unseen data because of overfitting.

(c) True Relationship Is Not Linear: Training RSS

If the true relationship is nonlinear, the cubic model can capture some of that nonlinearity. But because the question concerns training RSS, the key fact remains that the cubic model contains the linear model.

Therefore, the cubic model will generally have a lower or equal training RSS than the linear model.

For training RSS, the cubic model will generally perform at least as well as the linear model because it includes the linear model as one of its possible forms.

(d) True Relationship Is Not Linear: Test RSS

This case depends on how strongly nonlinear the true relationship is and how much noise is present.

If the relationship is only slightly nonlinear, the linear model may perform almost as well as the cubic model and could even have a lower test RSS because it has lower variance. If the relationship is strongly nonlinear, the cubic model may perform much better because it can capture the nonlinear pattern.

Therefore, without more information about the true relationship and noise level, we cannot determine which model will have the lower test RSS.

When the true relationship is nonlinear, the cubic model may be better, but not always. If the nonlinearity is weak, the simpler linear model may generalize better. If it is strong, the cubic model may have an advantage.